Candidate Name: ______________________ Class: ______________

PIONEER JUNIOR COLLEGE
While I Live, I Learn

JC2 PRELIMINARY EXAMINATION
Higher 2

MATHEMATICS **9740/01**
Paper 1 **16 September 2014**
3 hours

Additional Materials: Cover page
Answer papers
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your full name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

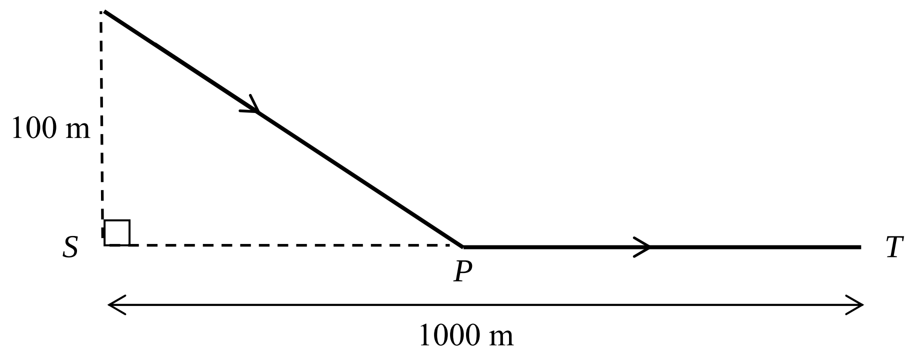
This document consists of **7** printed pages and **1** blank page.

1 Angeline, Benjamin and Clement are 3 tenants of a house, and they have agreed to share the monthly utility bill comprising of 3 components, namely electricity, water and gas, according to their usage patterns. The percentages of share that each tenant has agreed to pay across the 3 utility components are given in the following table:

	Percentage of share across 3 components		
	electricity	water	gas
Angeline	40	60	0
Benjamin	50	30	20
Clement	70	20	10

In a particular month, they received a utility bill with a total amount of \$400 and the breakdown by percentages of cost for electricity, water and gas is 50%, 40% and 10% respectively. Write down and solve equations to find the share of the bill for each of the tenants for that month. [4]

2



A man is controlling a toy plane so that it will begin to descend when it is 100 m directly above a point S on the ground. The plane will descend in a straight line at a constant speed of $20\ \text{ms}^{-1}$ till it lands at a point P and will travel in a straight line at a constant speed of $40\ \text{ms}^{-1}$ to a point T, which is 1000 m from S, as shown in the above diagram. If the man has to bring the toy plane to reach T as soon as possible, find the gradient of the slope at which the plane should descend. [6]

3 Relative to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $3\mathbf{a}$ respectively. Point P lies on AB such that $AP:PB=1:2$ and point Q lies on OP produced such that $5OQ=9OP$.

(i) Find the position vector of Q in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that Q lies on BC and hence state the ratio $BQ:QC$. [3]

(iii) Show that the area of triangle APQ is $k\left|\mathbf{a}\times\mathbf{b}\right|$, where k is a constant to be determined. [2]

4 A sequence is defined by $u_{n+1}=u_n+\dfrac{2n^2-1}{4n^2-1}$ and $u_1=1$.

(i) Write the values of u_2, u_3 and u_4. Hence find a conjecture for u_n in terms of n. [3]

(ii) Prove by mathematical induction the conjecture formed in part **(i)**. [4]

(iii) State, with a reason, whether the sequence is convergent. [1]

5 The variables x, y and z are connected by the following differential equations.

$$(2+y)^2\frac{\mathrm{d}^2z}{\mathrm{d}y^2}+1=0 \qquad \text{(A)}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\mathrm{e}^z \qquad \text{(B)}$$

(i) Given that $y>-2$ and that when $y=-1$, $\dfrac{\mathrm{d}z}{\mathrm{d}y}=1$ and $z=0$, solve equation (A) to find z in terms of y. [3]

(ii) Hence find y in terms of x. [3]

(iii) The result in part **(ii)** represents a family of curves. On a single diagram, sketch two non-linear members of the family of curves, showing clearly any asymptote. Write down the equations of these graphs. [2]
[You need not find axial intercepts or points of intersection, if any.]

6 Two different cars, A and B, are being made to clock distances at a constant speed to test for their fuel efficiency.

(i) For the first kilometre, car A used 125 ml of petrol and the amount of petrol used for each subsequent kilometre clocked forms a geometric progression. By the 20^{th} kilometre clocked, car A has used 1000 ml of petrol. Show that the total amount of petrol used after the 20^{th} kilometre can never exceed 98 ml. [4]

For car B, readings are being taken differently. At each kilometre clocked, the total amount of petrol used by car B from the beginning of the test, is being recorded. The tester noticed that the readings on and after the 20^{th} kilometre follow an arithmetic progression, with a common difference d.

(ii) Assuming car B used the same amount of petrol as that of car A for the first 20 kilometres and that by the time car B clocked 24 kilometres, it has used 1020 ml of petrol, find d. [2]

(iii) Which car has better fuel efficiency for the 24^{th} kilometre? [2]

7 It is given that

$$\mathrm{f}(x) = \begin{cases} \dfrac{1}{1+\sqrt{x}} & 0 \le x < a^2, \\ -2 & a^2 \le x < 5, \end{cases}$$

and that $\mathrm{f}(x+5) = \mathrm{f}(x)$ for all real values of x, where $0 < a < \sqrt{5}$.

(i) Show that $\mathrm{f}(100) = 1$. [1]

(ii) Sketch the graph of $y = \mathrm{f}(x)$ for $-5 + a^2 \le x < 5 + a^2$. [3]

(iii) By using the substitution $u = 1 + \sqrt{x}$, find

$$\int_0^{a^2} \mathrm{f}(x)\, \mathrm{d}x,$$

in terms of a. [5]

8 **(i)** Without using a calculator, show that

$$\left(1-\sqrt{3}\mathrm{i}\right)^{-12}=\frac{1}{4096}. \qquad [2]$$

(ii) Hence solve the equation

$$z^6\left(1-\sqrt{3}\mathrm{i}\right)^{12}-1=0,$$

giving the roots in the form $r\mathrm{e}^{\mathrm{i}\theta}$, where $r>0$ and $-\pi<\theta\le\pi$. Show the roots on an Argand diagram. [4]

(iii) Show that $\left(z-r\mathrm{e}^{\mathrm{i}\theta}\right)\left(z-r\mathrm{e}^{-\mathrm{i}\theta}\right)=z^2-2rz\cos\theta+r^2$. [2]

(iv) Use your answers in parts **(ii)** and **(iii)** to express $z^6\left(1-\sqrt{3}\mathrm{i}\right)^{12}-1$ as the product of three quadratic factors with real coefficients, giving each factor in non-trigonometrical form. [3]

9 The function f is defined by

$$\mathrm{f}: x \mapsto x^2-2\lambda x-3\lambda^2, \quad x\in\mathbb{R},$$

where λ is a positive constant.

(i) Sketch the curve of $y=\mathrm{f}(x)$. [1]

(ii) If the domain of f is restricted to $x\ge k$, state the least value of *k*, in terms λ for which the function f^{-1} exists. [1]

In the rest of the question, the domain of f is defined as in part **(ii)**.

Given that the value of λ is 1,

(iii) find f^{-1}, stating its domain, [3]

(iv) show that the solution of the equation $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$ satisfies the equation $x^2-2x=x+3$. Hence, find the exact solution of the equation $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$. [3]

Describe fully a sequence of transformations which will transform the curve $y=x^2$ on to the curve in part **(i)** with $\lambda=1$. [3]

10 The curve C has equation

$$y = \frac{px+q}{x^2+r},$$

where p, q and r are constants.

It is given that C has an asymptote at $x = 2$ and a turning point at $\left(4, \frac{1}{4}\right)$.

(i) Find the values of p, q and r. [4]

(ii) Sketch C, stating clearly equations of asymptotes, and the coordinates of turning points and intersections with the axes. [3]

(iii) Hence, solve the inequality

$$\left|\frac{px+q}{x^2+r}\right| \le 1.$$ [3]

11 A curve C has parametric equations

$$x = \frac{2}{(1-t)^2}, \quad y = \frac{t}{1-t} \qquad \text{where } t \ne 1.$$

(i) Points P and Q on C have parameters p and q respectively such that $p < q$. Point A is a point on C when $t = 0$. The normal to C at point P and the normal to C at point Q both pass through point A. Find p and q. [5]

(ii) Find the area of the region bounded by C for $y \ge 0$, the line $x = 8$ and the x-axis. [3]

12 The planes p_1 and p_2, which meet in the line l, have vector equations

$$\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 7\mathbf{k} + s(2\mathbf{i} + 3\mathbf{k}) + t(-4\mathbf{j} + 5\mathbf{k}) \qquad \text{and} \qquad \mathbf{r} \bullet \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} = 24,$$

respectively.

(i) Find a vector equation of l. [4]

(ii) Find the cartesian equation of the plane p_3 which contains l and the point $(3, -4, 1)$. [3]

(iii) Using parts **(i)** and **(ii)**, state with a reason, the number of solutions for the following system of equations

$$\begin{aligned} 6x - 5y - 4z &= -32 \\ 5x - y + 3z &= 24 \\ 9x - 2y + 5z &= 40. \end{aligned}$$

[3]